



PROJECT ON

Movie Sales Analysis

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AGENDA

BUSINESS PROBLEM

PROJECT OVERVIEW

DATA PREPARATION

DATA VISUALISATION

APPLICATIONS

RECOMMENDATIONS

CHALLENGES

CONCLUSION

Q/A



BUSINESS PROBLEMS

Background

- The global film industry generates billions of dollars every year. Yet most movies fail to recover their production costs. Studios invest millions without fully understanding what drives success.

The Problem

- Stakeholders struggle to answer:
- Which genres generate the highest revenue?
- Which directors deliver the best returns?
- Does a high rating always mean high revenue?
- What is the ideal runtime for maximum earnings?
- Which years were most profitable?

PROJECT OVERVIEW

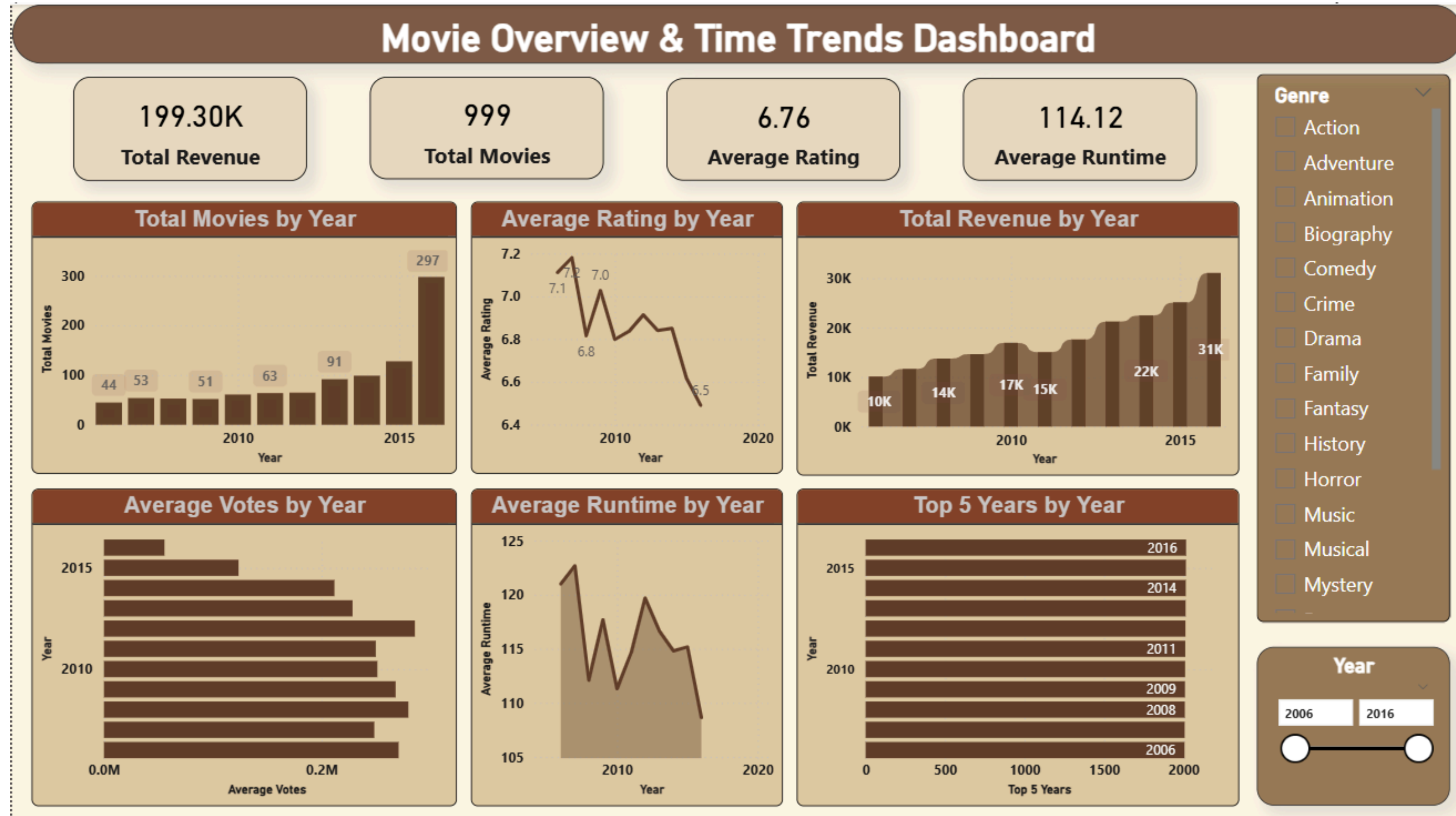
- Analyzed movie performance patterns using Power BI
- Identified top performing genres, directors, and movies
- Evaluated revenue trends and box office performance over the years
- Built interactive dashboards for insights and decision making
- Enabled studios and investors to make smarter production decisions

DATA PREPARATION

- Loaded raw IMDb CSV data using Power BI Get Data
- Replaced NULL values in Revenue with 0
- Replaced NULL values in Metascore with 0
- Extracted Genre from multi-genre column using Power Query
- Created Rating Category column — Excellent, Good, Average, Below Average
- Verified all column data types — numeric, text and decimal
- Created DAX measures for cards and calculations across all pages

DATA VISUALISATION

I. Movie Overview & Time Trends

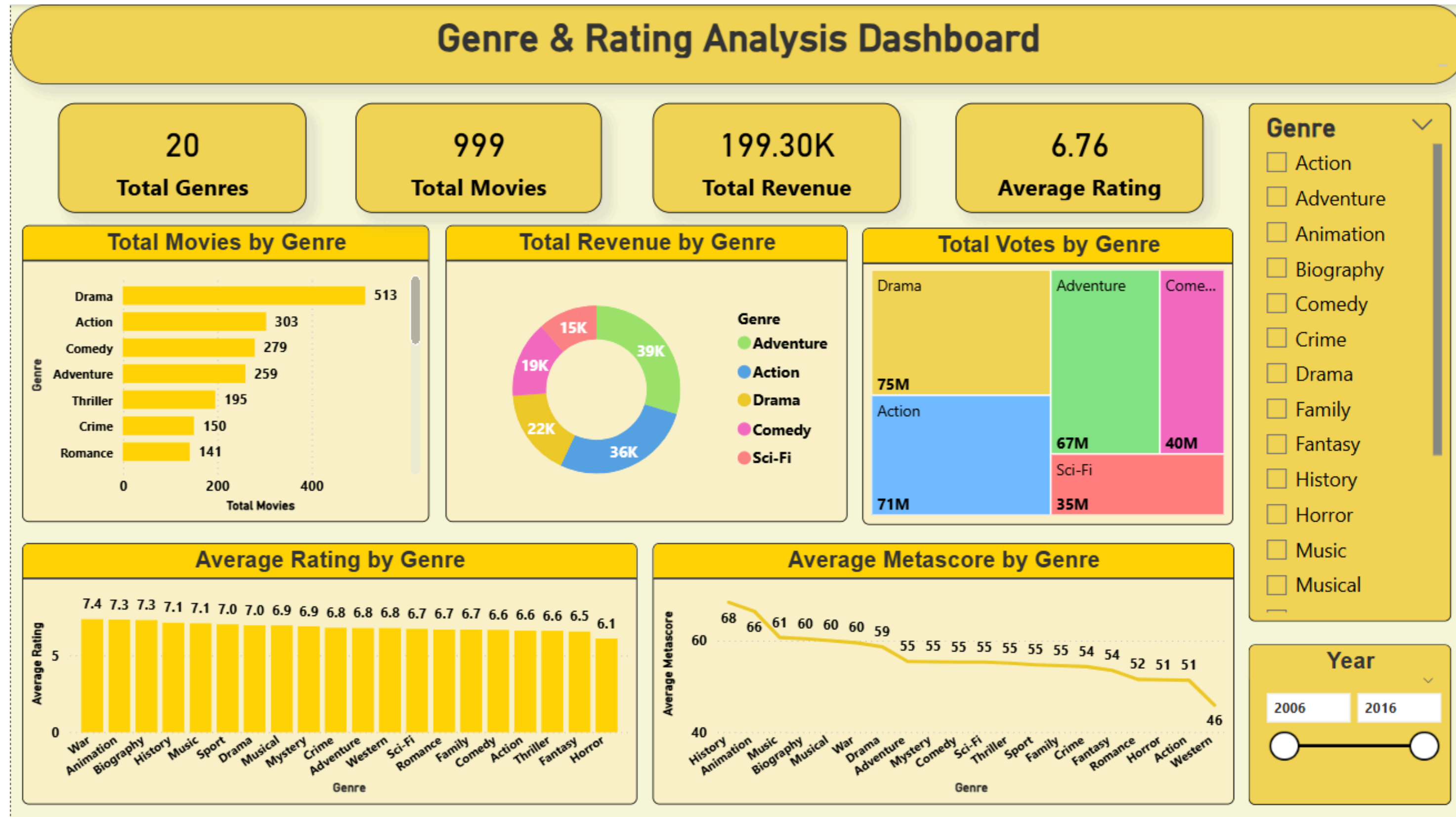


KEY INSIGHTS

- Total of 1000 movies analyzed spanning multiple years
- Movie production has steadily increased over the years
- Total box office revenue shows an overall upward trend
- Average IMDb rating has remained relatively stable around 6.7
- 2016 recorded the highest number of movie releases
- Average movie runtime is approximately 113 minutes
- Recent years show higher vote counts indicating growing audience engagement

DATA VISUALISATION

2. Genre & Rating Analysis

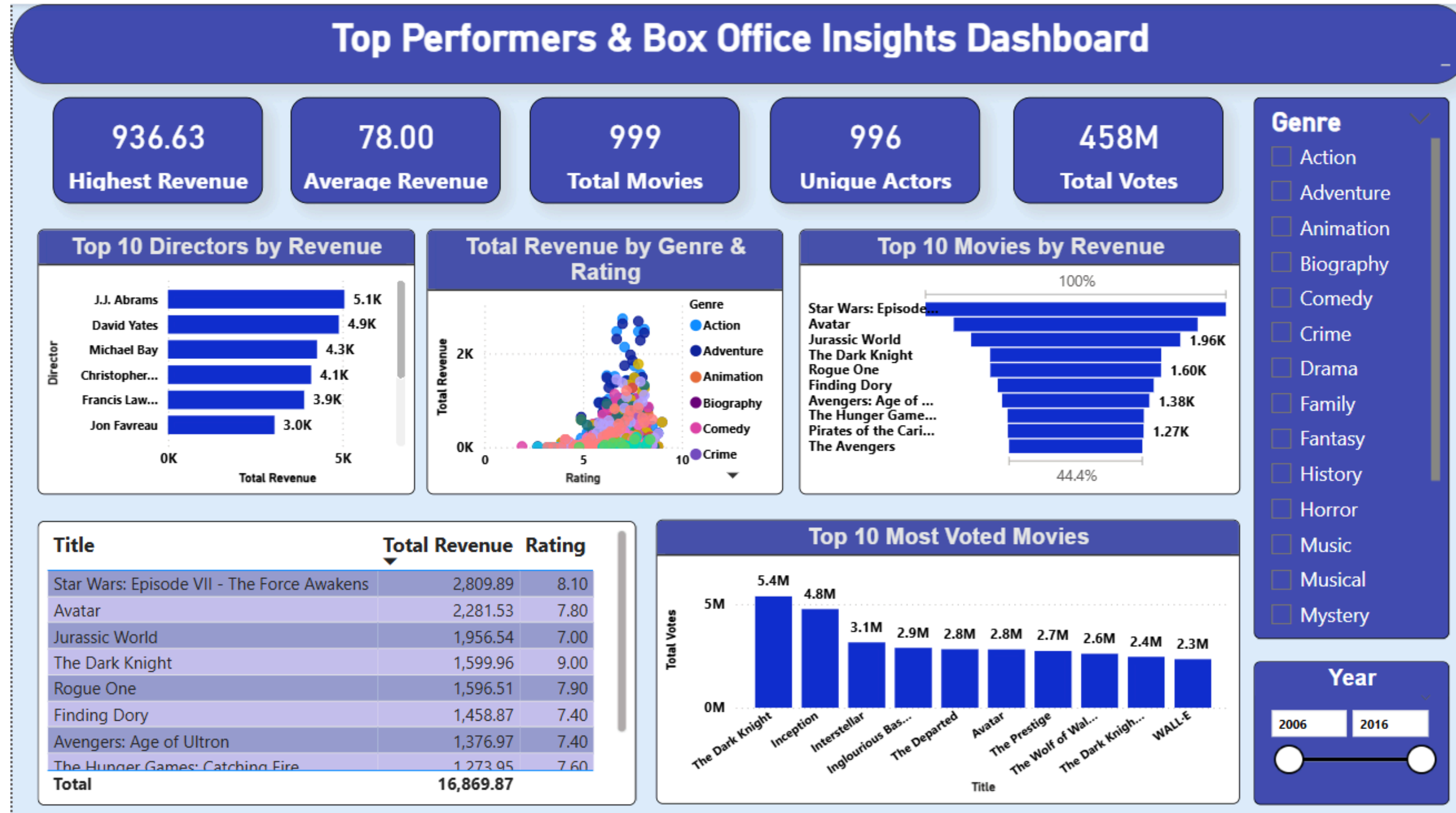


KEY INSIGHTS

- Action and Adventure genres generate the highest total revenue
- Drama is the most commonly made genre but does not top revenue
- Biography and Animation genres receive the highest IMDb ratings
- Action genres receive the highest number of audience votes
- Critic scores and audience scores often disagree — especially for Action movies
- Horror genre shows lower Metascores but strong audience engagement

DATA VISUALISATION

3. Top Performers & Box Office Insights

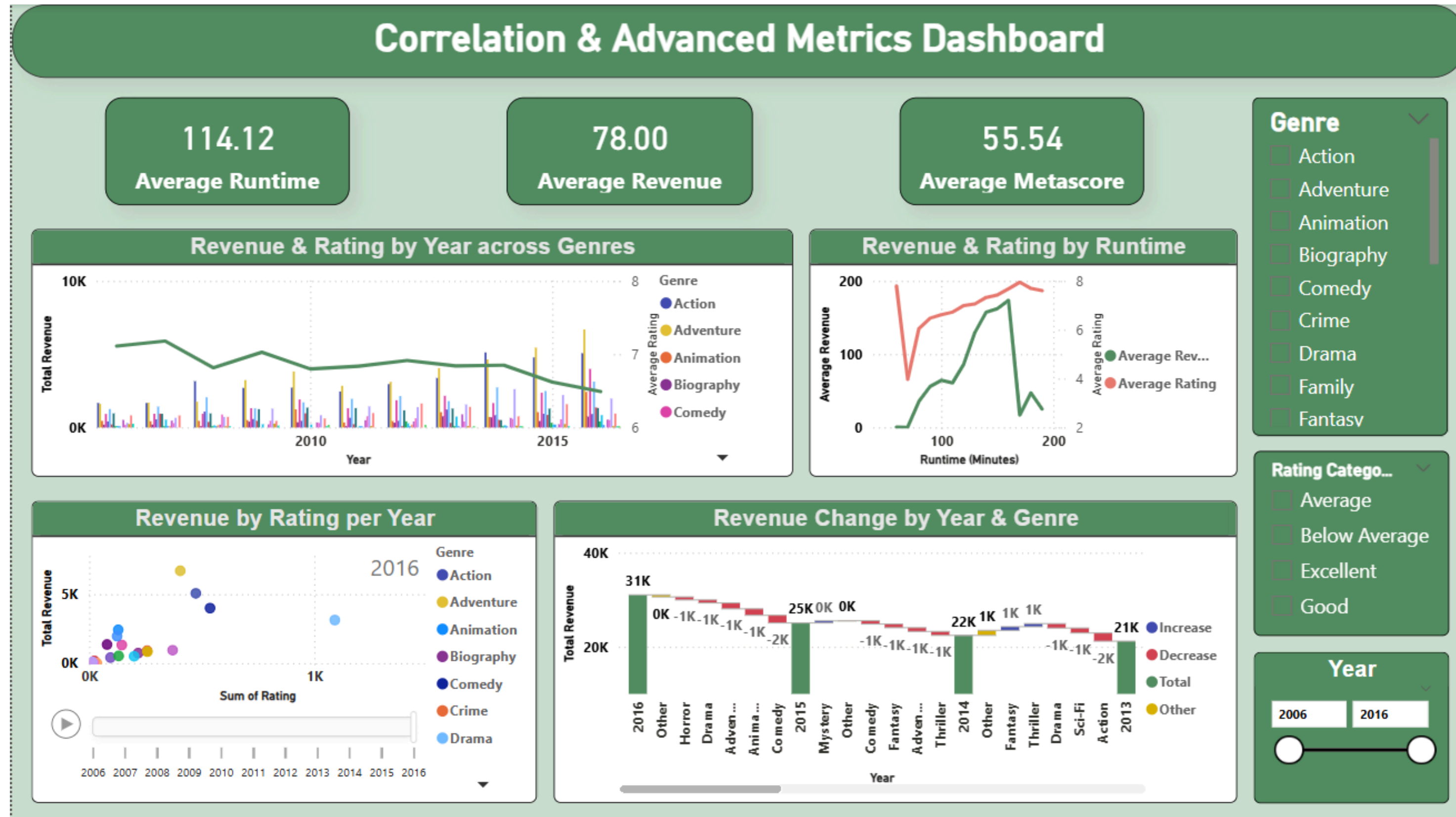


KEY INSIGHTS

- A small group of directors generates a disproportionate share of total revenue
- The top 2-3 movies earn dramatically more than the rest
- High votes does not always mean high revenue
- High ratings do not always guarantee high box office earnings
- Franchise and action movies dominate both revenue and vote counts
- Avg revenue per movie is approximately \$72 million

DATA VISUALISATION

4. Correlation & Advanced Metrics



KEY INSIGHTS

- Movies with runtime between 100–130 minutes perform best in revenue
- Higher rated movies do not always generate higher revenue
- Action and Adventure genres consistently outperform others every year
- Revenue has grown significantly from 2006 to 2016
- Certain years show sharp revenue drops indicating market downturns
- The relationship between rating and revenue varies strongly by genre

APPLICATION

- Identify high revenue potential genres before investing in production
- Support data driven casting and director selection decisions
- Help streaming platforms decide which movies to acquire or produce
- Enable marketing teams to focus budgets on high engagement movies
- Assist investors in understanding revenue patterns and risks
- Provide studios with insights to time movie releases for maximum impact

RECOMMENDATION

- Invest more in Action and Adventure genres for higher revenue returns
- Balance critic quality and commercial appeal when selecting projects
- Target the 100–130 minute runtime sweet spot for best performance
- Prioritize directors with proven high revenue track records
- Use audience vote counts as an early signal of movie popularity
- Diversify genre portfolio to reduce dependency on single genre performance

CHALLENGES

- Missing Revenue values required careful handling without deleting records
- Genre column had multiple genres per movie requiring extraction of Genre
- DAX table name had (l) suffix causing measure errors that needed debugging
- Balancing all 20 analysis questions across 4 dashboard pages within limited space

CONCLUSION

- Movie commercial success is driven by genre, director, and audience engagement
- High IMDb ratings alone do not guarantee box office revenue
- A small number of directors and genres dominate industry earnings
- Data driven analysis can significantly improve production investment decisions
- Power BI dashboards make complex movie data accessible and actionable
- Studios and investors can use these insights to reduce risk and maximize returns

**THANK
YOU**



